INTRODUCTION TO DATABASE / RDBMS

OBJECTIVES

- File-Based vs Database
- Characteristics of Database Approach
- Actors
- Advantages and disadvantages of DBMS
- Brief History of RDBMS

TRADITIONAL FILE-BASED

- manual file is set up to hold all external and internal correspondence relating to a project, product, employee, or client, such files are labeled and stored in cabinets
- for security, cabinets have locks or may be located in secure areas of the building

FILE-BASED SYSTEM

- collection of application programs that perform services for the end-users such as the production of reports
- an early attempt to computerize the manual filing system
- was developed in response to the needs of industry for more efficient data access

LIMITATIONS OF FILE-BASED APPROACH

- Separation and isolation of data
- Duplication of data
- Data dependence
- Incompatible file formats
- Fixed queries / proliferation (production or creation) of application programs

DATABASE

- a shared collection of logically related data, and description of this data, designed to meet the information needs of an organization

DATABASE MANAGEMENT SYSTEM (DBMS)

- a software system that enables users to define, create, maintain, and control access to the database

CHARACTERISTICS OF DATABASE APPROACH

1. **Self-describing nature of dbase system**

- The database contains not only the dbase itself but also a complete definition or description of the dbase structure and constraints. The definition is stored in the system catalog (also called as meta-data).

CHARACTERISTICS OF DATABASE APPROACH

- **2. Insulation between programs and data, and data abstraction**
- The structure of data files is stored in the DBMS catalog separately from the access programs (program independence). In DBMS environment, in adding another piece of data, we just need to change the description of data in meta-data, no programs are changed.

  Data abstraction – characteristic that allows program-data independence and program operation independence.

CHARACTERISTICS OF DATABASE APPROACH

- **3. Support of multiple views of the data**
- A database typically has many users, each of whom may require different perspective or view of the dbase.

  View – may be a subset of dbase

CHARACTERISTICS OF DATABASE APPROACH

4. Sharing of data and multiuser transaction processing

- Must allow multiple users to access the dbase at the same time
- Concurrency control – the DBMS must include software to ensure that several users trying to update the same data do so in a controlled manner so that the result of the updates is correct
- On-Line Transaction Processing (OLTP) – DBMS should ensure that each subset or record can be accessed by only one user at a time

ACTORS ON THE SCENE

- Database Administrators (DBA) – administers database environment such as database itself and DBMS and related software.
- DBA is responsible for authorizing access to the dbase, for coordinating and monitoring its use, and for acquiring software and hardware resources.
- accountable for problems such as breach of security or poor system response time

ACTORS ON THE SCENE

- **Database Designers** – responsible for identifying the data to be stored in the dbase and for choosing appropriate structures to represent and store this data
- **End-users** – people whose jobs require access to the dbase for querying, updating, and generating reports

ACTORS ON THE SCENE

Categories of End-Users

1. **Casual end user** – occasionally access the dbase, but they need different information each time
    - typically middle- or high-level managers or other occasional browsers
2. **Naïve or parametic end users** – make up a sizeable potion of dbase end user
    - their main job function revolves around constantly querying and updating the dbase
    - example: bank tellers, reservation clerks for airlines, hotels and car rentals

ACTORS ON THE SCENE

Categories of End-Users

3. Sophisticated end users – include engineers, scientists, business analysts, and others who thoroughly familiarize themselves with the facilities of DBMS

4. Stand-alone users – maintain personal dbases by using ready-made program packages that provide easy-to-use menu- or graphics-based interfaces

ACTORS ON THE SCENE

- **System analysts and application programmer (software engineers)**

  1. System analysts – determine the requirements of end-users and develop specifications for canned transactions (using standard types of queries and updates)

  2. Application programmers – implement these specifications as programs, then they test, debug, document and maintain the canned transactions

A BRIEF HISTORY OF RDBMS

- Dr. Edgar F. Codd pioneered the concept of relational database model in 1970.
- Dr. Codd presented his relational database theory when he published his seminal paper entitled “A Relational Model of Data for Large Shared Data Banks”, with the Communications of the Association of Computing Machinery (ACM) in June 1970
- In mid-1970’s, IBM corporation (then the world’s biggest computer company), first implemented
  the Structured Query Language (SQL) on their research project conducted, known as System R.

A BRIEF HISTORY OF RDBMS

- SQL is the standard language to access relational databases and was a ground-breaking invention of Dr. Codd at the IBM Research Labs in 1971
- Based on historical legends, IBM was not able to commercialize it due to reasons of internal conflict with their other successful database products during that time.
- This is how Oracle Corporation made it first to market the SQL software in 1979. Well, that SQL product is now the world's renowned ORACLE Database

RDBMS

- a collection of interrelated *information* that is organized into *tables*.
- We store data in each table by *rows*, and the data is arranged into *columns*.
- We store tables in database *schemas*.
  - *schema* is a logical container for the database objects that the user creates
- In relational database, we can grant permissions to other users so that they can access our tables, or the other users can grant us permission to access their tables.
- In other words, we can share data to others easily.